

# Passive Thermal Design of a 6U Small Satellite Under a High-Beta Bounding Hot Case

Runze Shi

*Master of Applied Science, Mechanical Engineering*

## Abstract

This report documents the preliminary passive thermal design and radiator sizing trade-off analysis of a high-power 6U technology-demonstration CubeSat in a circular 600 km dawn-dusk Sun-Synchronous Orbit (SSO). In dawn-dusk SSO configurations, high solar beta angles during solstices drive near-continuous solar exposure, presenting extreme over-temperature challenges for internal avionics and battery packs. We present **Passive Redesign v2**, which eliminates the need for active cooling or deployable mechanisms. The redesign uses passive thermal zoning, surface coating selection (AZ-93 silicate paint and silvered FEP tape), and orbit-attitude-dependent radiator placement on the anti-Sun cross-track face. A dual-axis system-level radiator sizing sweep reveals that a 0.02 m<sup>2</sup> radiator offers the optimal balance between hot-case thermal safety margin (avionics peak at 39.2°C) and cold-case heater power penalties (5.92 Wh/orbit).

## 1 Introduction

Modern small satellites are increasingly packed with high-power payloads and avionics, leading to significant internal power dissipation. A standard 6U CubeSat (10 cm × 20 cm × 30 cm) technology-demonstrator requires robust thermal architecture to operate under bounding orbital conditions.

This study focuses on a 6U spacecraft in a circular 600 km dawn-dusk SSO with an inclination of  $\approx 97.8^\circ$ . Under solstice conditions, the orbit plane tilts relative to the sun vector, yielding a high beta angle ( $\beta = 74^\circ$ ) that exceeds the critical shadow angle ( $\approx 66.07^\circ$ ), causing the satellite to remain in continuous sunlight. The SSO geometry and sunlit schedule are shown in Figure 1.

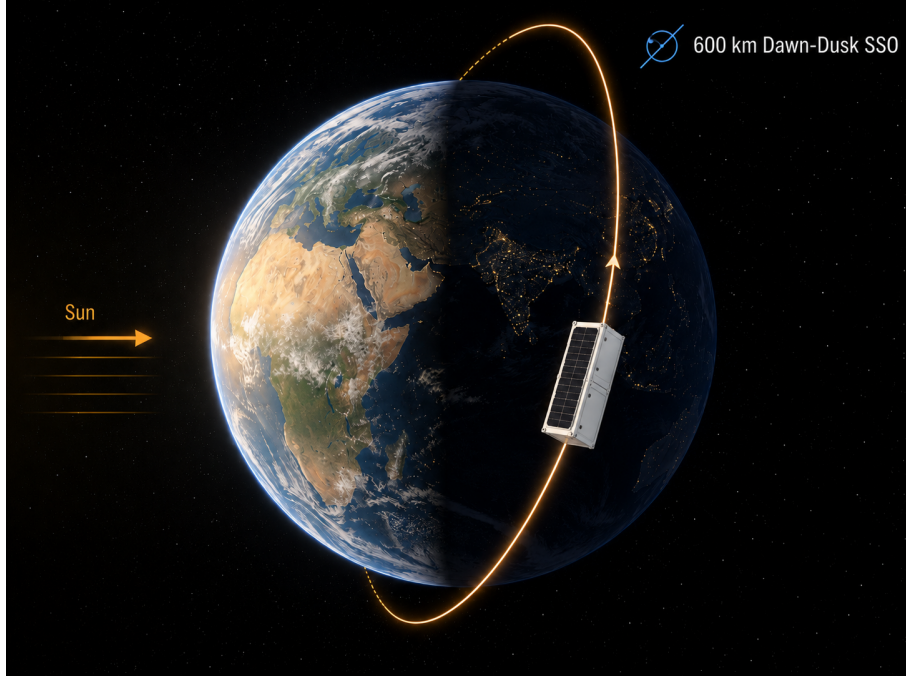


Figure 1: Circular 600 km Dawn-Dusk SSO Orbit Geometry, illustrating the permanent solar illumination vector on the cross-track panel.

## 2 Thermal Limits and Component Specifications

Thermal node mass, capacitance, and operational limits are traceable to engineering standards. The component details are listed in Table 1.

Table 1: Component Mass, Capacitance, and Temperature Limits

| Node ID        | Mass [kg] | $c_p$ [J/kg·K] | Min Limit [°C] | Max Limit [°C] | Internal Diss. [W] |
|----------------|-----------|----------------|----------------|----------------|--------------------|
| Battery        | 1.5       | 800.0          | -10.0          | 45.0           | 0.5                |
| Avionics       | 1.0       | 900.0          | -20.0          | 60.0           | 10.0 – 15.0        |
| Payload        | 2.0       | 900.0          | -10.0          | 50.0           | 0.0 – 15.0         |
| Structure      | 3.0       | 900.0          | -40.0          | 80.0           | 0.0                |
| Radiator       | 0.5       | 900.0          | -100.0         | 100.0          | 0.0                |
| External Shell | 1.5       | 900.0          | -100.0         | 100.0          | 0.0                |

The 6-node lumped-parameter thermal network architecture, conductive couplings  $G_{ij}$ , and external load interfaces are shown schematically in Figure 2.

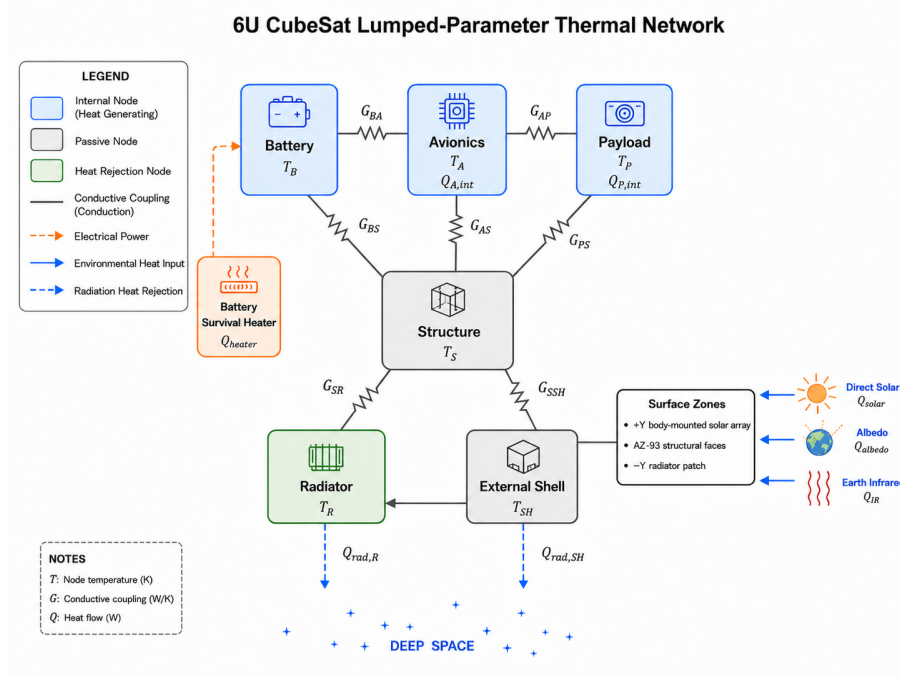


Figure 2: 6-Node Lumped Parameter Thermal Network diagram, showing inter-node conductive paths and environmental radiation links.

### 3 Material Library and Optical Properties

All thermo-optical coatings used in this design are canonical candidates sourced directly from the NASA SmallSat State of the Art (SoA) Thermal Control webpage, specifically Tables 7-3 and 7-4. The material properties catalog is summarized in Table 2.

Table 2: Material Library and BOL Optical Properties

| Material ID           | Intended Role        | $\alpha_s$ (BOL) | $\epsilon_{IR}$ | NASA Source | Concise Description                  |
|-----------------------|----------------------|------------------|-----------------|-------------|--------------------------------------|
| AZ-93 Silicate        | External shell       | 0.16             | 0.90            | Table 7-3   | Inorganic white thermal paint        |
| 0.005" FEP/Ag/Inconel | Primary radiator     | 0.07             | 0.79            | Table 7-4   | Second-surface silvered FEP tape     |
| VDA/200HN/PSA         | Reference insulation | 0.08             | 0.03            | Table 7-4   | Metallized tape, very low emissivity |
| Z306 Polyurethane     | Comparison paint     | 0.92             | 0.89            | Table 7-3   | Matte black polyurethane paint       |
| solar_cell_assembly   | Solar panels         | 0.73             | 0.82            | Section 7.1 | Body-mounted solar cells             |

### 4 Baseline Failure Analysis

The initial baseline configuration mapped high-absorptance black paint ( $\alpha = 0.92, \epsilon = 0.89$ ) to several outer shell faces representing solar panel substrates. Under the bounding hot case ( $\beta = 74^\circ$ ), the large surface absorption caused extreme solar heat loading. This resulted in catastrophic thermal limits breach across multiple nodes (Battery peaking at  $80.6^\circ\text{C}$ , Avionics peaking at  $81.6^\circ\text{C}$ ).

Additionally, using a low-emissivity material (like metallized VDA tape,  $\epsilon = 0.03$ ) as a radiator was shown to cause thermal traps, raising avionics to  $104.1^\circ\text{C}$  due to the inability to reject heat to deep space.

## 5 Passive Redesign v2 Architecture

To resolve these violations passively, we implemented three key solutions in Redesign v2:

- **Thermal Surface Zoning:** We mapped AZ-93 white paint ( $\alpha = 0.16, \epsilon = 0.90$ ) to five non-power shell faces to minimize solar absorption while maximizing deep-space emittance.
- **Radiator Face Orientation Physics:** In dawn-dusk SSO, the cross-track  $+Y$  face is permanently Sun-facing, while the opposite  $-Y$  face is permanently shaded (anti-Sun). The dedicated radiator ( $0.02 \text{ m}^2$  silver FEP tape,  $\alpha = 0.07, \epsilon = 0.79$ ) was relocated to the anti-Sun  $-Y$  face to achieve the lowest integrated environmental heat load.
- **Solar Cell Electrical Offset:** Since 18% of the absorbed solar energy on the solar arrays is converted to electricity and exported out of the thermal network, we applied an electrical conversion offset:

$$\alpha_{\text{eff}} = \alpha_s - \eta_{\text{elec}} = 0.73 - 0.18 = 0.55 \quad (1)$$

Figure 3 illustrates the surface coating allocations and zoning layout for the redesigned spacecraft.

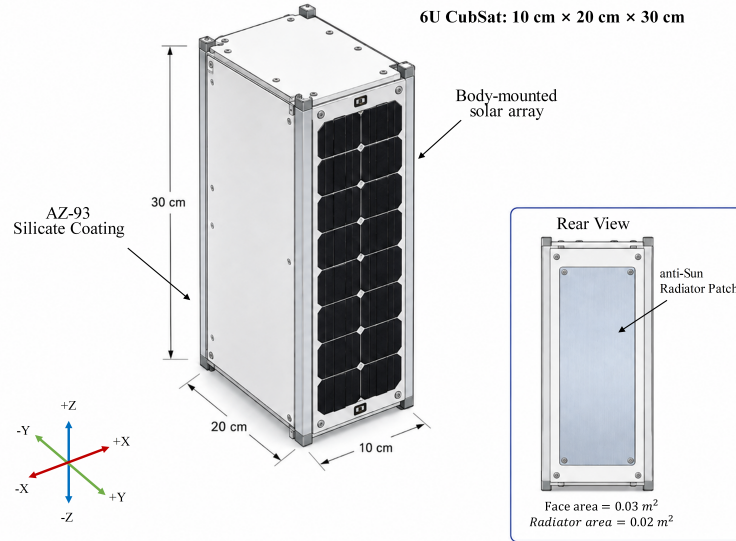


Figure 3: Spacecraft Passive Thermal Zoning layout (Redesign v2), showing body-mounted solar cell arrays on the Sun-facing  $+Y$  face, AZ-93 white paint on the general shell, and the silver FEP radiator on the anti-Sun  $-Y$  face.

## 6 Simulation Results

Applying the Redesign v2 architecture, all scenarios successfully converged to PASS conditions. Table 3 summarizes the steady-state node temperatures.



Table 3: Steady-State Temperature Ranges under Redesign v2

| Node ID       | Nominal ( $\beta = 65^\circ$ ) [ $^\circ\text{C}$ ] | Bounding Hot ( $\beta = 74^\circ$ ) [ $^\circ\text{C}$ ] | Cold Case ( $\beta = 60^\circ$ ) [ $^\circ\text{C}$ ] |
|---------------|-----------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------|
| Battery       | 14.8 to 15.1                                        | 35.0 to 35.4                                             | 5.0 to 10.0                                           |
| Avionics      | 16.4 to 17.1                                        | 38.1 to 39.2                                             | -0.8 to 0.8                                           |
| Payload       | 14.1 to 17.7                                        | 34.0 to 40.3                                             | -1.7 to -0.5                                          |
| Structure     | 13.2 to 14.2                                        | 33.5 to 34.8                                             | -2.2 to -0.1                                          |
| Heater Duty   | 0.0%                                                | 0.0%                                                     | 36.7%                                                 |
| Heater Energy | 0.0 Wh                                              | 0.0 Wh                                                   | 5.92 Wh/orbit                                         |

## 7 Radiator Sizing System-Level Trade-Off

A sweep of radiator area  $A_{\text{rad}}$  from 0.02 to 0.06 m<sup>2</sup> was performed across all three scenarios to evaluate the trade-off between hot-case avionics peak temperature and cold-case heater power draw (Table 4).

Table 4: Radiator Area Sweep Trade-Off Results

| Area [m <sup>2</sup> ] | Hot Peak Avionics [ $^\circ\text{C}$ ] | Cold Min Battery [ $^\circ\text{C}$ ] | Cold Heater Duty [%] | Cold Heater Energy [Wh] |
|------------------------|----------------------------------------|---------------------------------------|----------------------|-------------------------|
| 0.02                   | 39.17                                  | 5.00                                  | 36.7%                | 5.92                    |
| 0.03                   | 34.87                                  | 5.00                                  | 41.5%                | 6.69                    |
| 0.04                   | 31.15                                  | 5.00                                  | 49.0%                | 7.90                    |
| 0.05                   | 27.80                                  | 5.00                                  | 63.6%                | 10.26                   |
| 0.06                   | 24.81                                  | 4.99                                  | 60.9%                | 9.81                    |

Increasing the radiator size significantly drops the hot-case peak temperature, but imposes a severe 73% heater energy penalty in the cold case. Due to strict power limitations on small satellites, the \*\*0.02 m<sup>2</sup> radiator size is selected as the optimum thermal design\*\*.

## 8 Visualizations and Design Analysis

The transient temperature response and parameter sizing trade-offs are visualized below. Figure 4 illustrates the nodal temperature histories under the continuous sunlight bounding hot case, showing stable convergence. Figure 5 depicts the dual-axis trade-off between peak hot avionics temperatures and cold battery margins across various radiator areas.

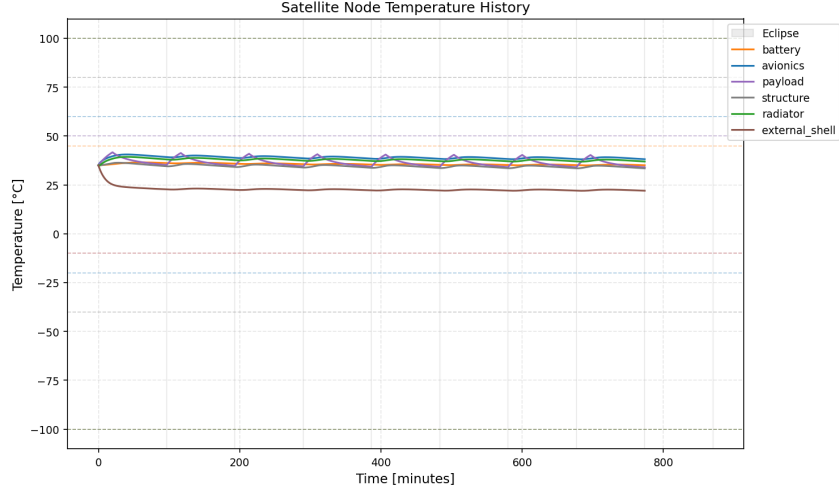


Figure 4: Nodal Temperature Histories Under Bounding Hot Case ( $\beta = 74^\circ$ , Continuous Sun-light Dawn-Dusk SSO Solstice).

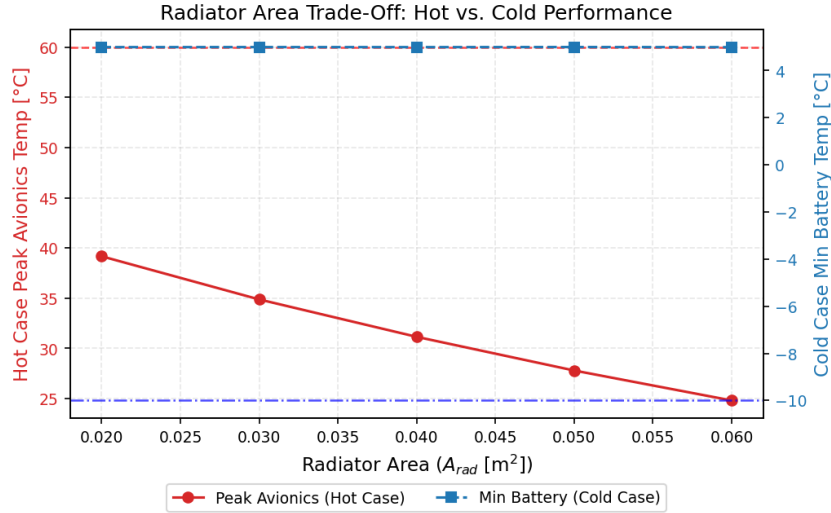


Figure 5: Dual-Axis Radiator Sizing Trade-Off (Hot Case Peak Avionics Temp vs. Cold Case Min Battery Temp).

## 9 Conclusion

This report demonstrates that a rigorous passive thermal redesign utilizing surface zoning, anti-Sun radiator placement, and electrical power offset calculations successfully resolves the hot-case over-temperature limits of a 6U CubeSat in 600 km dawn-dusk SSO. Sizing trades highlight the importance of balancing radiator design to minimize cold-case heater power penalties.